

Image Filter App

Description

A web-based application built with Streamlit that allows users to upload images and apply various filters in real-time. The app provides a side-by-side comparison of the original and filtered images with download capability.

Features

- **7 Image Filters:**
 - Original (no filter)
 - Black & White
 - Brightness adjustment
 - Artistic style
 - HDR effect
 - Blur
 - Sharpen
- **Interactive Controls:** Adjustable parameters for each filter
- **Image Comparison:** Slider to compare before/after results
- **Download:** Save filtered images in PNG format
- **Responsive UI:** Beautiful gradient background with animated effects

Technologies Used

- Python
- OpenCV (for image processing)
- Streamlit (web framework)
- NumPy (array operations)
- PIL/Pillow (image handling)

Installation & Usage

1. Install dependencies:

```
pip install streamlit opencv-python numpy pillow streamlit-image-comparison
```